

## DRAMMEN BERSKOG, Norway

WMO: 014800

Lat: 59.7536N

Lon: 10.1236E

Elev: 26

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 0.9        | 5.6 | -4.1                          | 4.4 | 1.9  | 0.9 | 5.8 | 7.7  | 17.2                  | 42.5 | 14.2 | 37.4 | 3.3                   | 330  | 0.541 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.9                   | 82.2            | 63.2 | 78.4 | 62.0 | 74.7 | 60.3 | 65.8                | 76.3 | 64.1 | 73.8 | 62.6 | 71.2 | 6.1                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.2                            | 83.9 | 68.8 | 60.6 | 79.0 | 67.2 | 58.8 | 74.1 | 66.1 | 30.6          | 76.4 | 29.3 | 73.9 | 28.2 | 71.1 | 71.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 13.5              | 11.4 | 9.8 | -1.3                       | 86.4 | 5.9                | 4.1 | -5.6                                               | 89.3 | -9.0       | 91.7 | -12.3      | 94.1 | -16.6      | 97.1 |
|                   |      |     | -1.8                       | 67.6 | 5.9                | 2.3 | -6.0                                               | 69.3 | -9.5       | 70.7 | -12.8      | 72.0 | -17.0      | 73.6 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 44.2   | 24.9  | 26.8  | 33.9  | 43.7  | 52.8  | 60.2  | 64.9  | 61.5  | 54.7  | 43.9  | 34.8  | 27.2  |
|                                                                   | DBStd       | 15.59  | 9.47  | 9.07  | 7.47  | 5.77  | 6.72  | 4.65  | 4.92  | 3.83  | 4.92  | 6.22  | 7.26  | 10.51 |
|                                                                   | HDD50       | 3562   | 778   | 649   | 498   | 205   | 44    | 0     | 0     | 0     | 16    | 208   | 457   | 707   |
|                                                                   | HDD65       | 7682   | 1243  | 1069  | 963   | 640   | 383   | 158   | 64    | 121   | 308   | 655   | 906   | 1172  |
|                                                                   | CDD50       | 1445   | 0     | 0     | 0     | 15    | 130   | 306   | 462   | 356   | 158   | 17    | 1     | 0     |
|                                                                   | CDD65       | 90     | 0     | 0     | 0     | 0     | 4     | 13    | 61    | 12    | 0     | 0     | 0     | 0     |
|                                                                   | CDH74       | 914    | 0     | 0     | 0     | 0     | 83    | 204   | 547   | 79    | 1     | 0     | 0     | 0     |
|                                                                   | CDH80       | 197    | 0     | 0     | 0     | 0     | 12    | 46    | 135   | 4     | 0     | 0     | 0     | 0     |
| Wind                                                              | WSAvg       | 4.3    | 4.4   | 4.2   | 4.3   | 4.6   | 4.6   | 4.9   | 4.5   | 4.0   | 4.2   | 3.8   | 3.8   | 4.0   |
| Precipitation                                                     | PrecAvg     | 33.4   | 2.4   | 1.7   | 1.8   | 1.9   | 2.4   | 3.1   | 3.2   | 3.6   | 3.5   | 3.7   | 3.7   | 2.3   |
|                                                                   | PrecMax     | 48.5   | 5.6   | 4.2   | 5.0   | 4.9   | 4.7   | 6.5   | 5.9   | 7.2   | 9.4   | 9.3   | 14.7  | 4.2   |
|                                                                   | PrecMin     | 25.5   | 0.6   | 0.4   | 0.1   | 0.4   | 0.4   | 0.5   | 0.4   | 1.0   | 1.3   | 0.9   | 0.6   | 0.6   |
|                                                                   | PrecStd     | 4.5    | 1.3   | 0.9   | 1.2   | 1.1   | 1.3   | 1.7   | 1.3   | 1.5   | 1.8   | 2.0   | 2.8   | 1.1   |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 47.0  | 48.4  | 57.9  | 68.3  | 81.5  | 85.4  | 87.7  | 80.0  | 72.1  | 62.7  | 55.9  | 51.3  |
|                                                                   |             | MCWB   | 41.7  | 40.1  | 45.2  | 52.3  | 60.5  | 62.2  | 66.1  | 62.3  | 59.8  | 54.2  | 50.4  | 45.6  |
|                                                                   | 2%          | DB     | 43.1  | 44.1  | 52.3  | 64.2  | 75.6  | 80.1  | 83.9  | 76.1  | 69.4  | 59.3  | 52.1  | 47.3  |
|                                                                   |             | MCWB   | 38.6  | 38.4  | 42.9  | 50.4  | 58.4  | 60.4  | 64.7  | 61.8  | 58.0  | 51.7  | 47.6  | 42.8  |
|                                                                   | 5%          | DB     | 40.2  | 41.2  | 48.0  | 59.8  | 71.0  | 75.0  | 80.4  | 73.2  | 66.8  | 56.7  | 48.6  | 44.2  |
|                                                                   |             | MCWB   | 36.6  | 36.8  | 39.6  | 47.6  | 55.9  | 58.8  | 63.6  | 60.6  | 56.7  | 51.2  | 45.1  | 40.6  |
|                                                                   | 10%         | DB     | 37.3  | 38.3  | 44.7  | 55.6  | 66.1  | 71.3  | 76.5  | 70.4  | 64.1  | 54.1  | 45.7  | 40.9  |
|                                                                   |             | MCWB   | 34.4  | 35.5  | 38.3  | 45.2  | 54.1  | 56.5  | 62.4  | 59.3  | 55.5  | 49.1  | 42.9  | 37.4  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 42.8  | 42.0  | 46.7  | 54.3  | 63.6  | 65.2  | 69.0  | 65.8  | 62.7  | 56.7  | 52.2  | 46.7  |
|                                                                   |             | MCDB   | 46.2  | 45.9  | 55.1  | 64.8  | 74.8  | 79.2  | 80.7  | 73.4  | 67.7  | 60.8  | 54.9  | 50.6  |
|                                                                   | 2%          | WB     | 39.6  | 39.2  | 43.4  | 51.1  | 60.9  | 62.8  | 67.1  | 64.2  | 60.3  | 54.1  | 48.3  | 43.3  |
|                                                                   |             | MCDB   | 42.3  | 42.4  | 50.3  | 61.6  | 70.8  | 73.7  | 78.8  | 71.2  | 66.4  | 57.5  | 51.3  | 46.6  |
|                                                                   | 5%          | WB     | 37.2  | 37.5  | 41.2  | 48.9  | 58.2  | 60.8  | 65.3  | 62.6  | 58.5  | 52.1  | 45.7  | 40.8  |
|                                                                   |             | MCDB   | 39.7  | 40.4  | 46.4  | 57.7  | 68.0  | 71.1  | 76.1  | 69.7  | 64.1  | 55.7  | 48.2  | 43.8  |
|                                                                   | 10%         | WB     | 34.6  | 35.6  | 38.9  | 46.6  | 55.5  | 58.9  | 63.7  | 61.2  | 57.1  | 49.9  | 43.1  | 37.6  |
|                                                                   |             | MCDB   | 36.9  | 38.1  | 43.8  | 54.1  | 63.2  | 67.9  | 73.5  | 67.8  | 62.1  | 53.1  | 45.4  | 40.4  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.6   | 11.1  | 15.8  | 18.4  | 18.5  | 18.4  | 17.9  | 16.7  | 16.0  | 12.7  | 9.6   | 10.0  |
|                                                                   |             | MCDBR  | 13.7  | 15.3  | 20.7  | 25.9  | 27.2  | 25.7  | 24.7  | 21.0  | 20.0  | 16.3  | 15.0  | 14.9  |
|                                                                   |             | MCWBR  | 10.5  | 10.3  | 12.6  | 13.9  | 12.5  | 11.2  | 9.6   | 9.5   | 11.0  | 10.5  | 11.5  | 11.8  |
|                                                                   | 5% WB       | MCDBR  | 12.9  | 13.1  | 19.4  | 23.1  | 22.7  | 21.0  | 21.3  | 16.9  | 17.2  | 13.6  | 13.6  | 14.2  |
|                                                                   |             | MCWBR  | 10.6  | 9.3   | 12.4  | 12.8  | 11.0  | 10.4  | 9.4   | 8.9   | 10.3  | 10.0  | 11.2  | 11.9  |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.258 | 0.298 | 0.324 | 0.368 | 0.366 | 0.355 | 0.376 | 0.369 | 0.337 | 0.323 | 0.280 | 0.232 |
|                                                                   | taud        |        | 2.218 | 2.295 | 2.332 | 2.325 | 2.416 | 2.464 | 2.426 | 2.440 | 2.518 | 2.475 | 2.311 | 2.134 |
|                                                                   | Ebn at Noon |        | 172   | 226   | 257   | 264   | 274   | 279   | 269   | 262   | 253   | 218   | 163   | 133   |
|                                                                   | Edh at Noon |        | 13    | 21    | 28    | 33    | 33    | 32    | 32    | 30    | 23    | 18    | 12    | 9     |
| All-Sky Solar Radiation                                           | RadAvg      |        | 97    | 295   | 773   | 1229  | 1554  | 1772  | 1571  | 1240  | 819   | 370   | 129   | 64    |
|                                                                   | RadStd      |        | 19    | 58    | 76    | 171   | 170   | 135   | 202   | 95    | 101   | 71    | 34    | 6     |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (53 neighbors) | N/A   | N/A     | +2.81  | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page